

ESERCIZIO 1:

a) $\forall \varepsilon > 0, \exists a \in \mathbb{R}$ t.c. $|f(x) - 3| < \varepsilon \quad \forall x < a$

b) **TEO (LAGRANGE)**: $f: [a, b] \rightarrow \mathbb{R}$, $f \in C^0([a, b])$, f DERIVABILE SU (a, b) .

ALLORA $\exists \hat{x} \in (a, b)$ t.c. $f'(\hat{x}) = \frac{f(b) - f(a)}{b - a}$

DIM: DEFINIAMO $g(x) = f(x) - \frac{f(b) - f(a)}{b - a} (x - a)$

$g \in C^0([a, b])$ E g È DERIVABILE SU (a, b)

$$g(a) = f(a) - \frac{f(b) - f(a)}{b - a} (\overset{0}{a - a}) = f(a)$$

$$g(b) = f(b) - \frac{f(b) - f(a)}{b - a} (\cancel{b - a}) = f(a)$$

SECONDO ROLLE $\exists \hat{x} \in (a, b)$ t.c. $g'(\hat{x}) = 0$ MA $g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$

QUINDI $g'(\hat{x}) = f'(\hat{x}) - \frac{f(b) - f(a)}{b - a} = 0 \Rightarrow f'(\hat{x}) = \frac{f(b) - f(a)}{b - a}$ ■

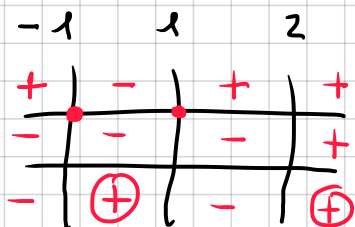
ESERCIZIO 2:

$$f(x) = \frac{x^2 - 1}{x - 2}$$

DOMINIO: $\Delta = \mathbb{R} \setminus \{2\}$

SIMMETRIA: $f(-x) = \frac{(-x)^2 - 1}{(-x) - 2} = \frac{x^2 - 1}{-x - 2}$ NÈ PARI NÈ DISPARI

SEGNO: $f(x) \geq 0$ ME $\frac{x^2 - 1}{x - 2} \geq 0$ $\begin{matrix} x^2 - 1 \geq 0 \\ x - 2 > 0 \end{matrix}$ $\begin{matrix} x \leq -1 \vee x \geq 1 \\ x > 2 \end{matrix}$



$$f(x) > 0 \quad -1 < x < 1 \vee x > 2$$

$$f(x) = 0 \quad x = \pm 1$$

$$f(x) < 0 \quad x < -1 \vee 1 < x < 2$$

LIMITI: $\lim_{x \rightarrow 2^-} \frac{x^2 - 1}{x - 2} = -\infty$ $\lim_{x \rightarrow 2^+} \frac{x^2 - 1}{x - 2} = +\infty$

$$\lim_{x \rightarrow +\infty} \frac{x^2 - 1}{x - 2} = \lim_{x \rightarrow +\infty} \frac{x^2 (1 - \frac{1}{x^2})}{x (1 - \frac{2}{x})} = \lim_{x \rightarrow +\infty} x = +\infty$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 - 1}{x - 2} = -\infty$$

ASINTOTI: NO ASINTOTI ORIZZONTALI, POSSIBILI ASINTOTI OBLIQUI.

$$\lim_{x \rightarrow \pm\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \frac{x^2 - 1}{x^2 - 2x} = \lim_{x \rightarrow +\infty} \frac{\cancel{x^2} (1 - \frac{1}{x^2})}{\cancel{x^2} (1 - \frac{2}{x})} = 1$$

$$\lim_{x \rightarrow +\infty} f(x) - x = \lim_{x \rightarrow +\infty} \frac{\cancel{x^2} - 1 - \cancel{x^2} + 2x}{x - 2} =$$

$$= \lim_{x \rightarrow \pm\infty} \frac{2x - 1}{x - 2} = \lim_{x \rightarrow +\infty} \frac{\cancel{x} (2 - \frac{1}{x})}{\cancel{x} (1 - \frac{2}{x})} = 2$$

$y = x + 2$ È L'EQUAZIONE DELL'ASINTOTO OBLIQUO PER $x \rightarrow \pm\infty$

DERIVATA PRIMA: $f'(x) = \left(\frac{x^2 - 1}{x - 2} \right)' = \frac{2x(x - 2) - (x^2 - 1)}{(x - 2)^2} =$

$$= \frac{\cancel{2}x^2 - 4x - \cancel{x^2} + 1}{(x - 2)^2} = \frac{x^2 - 4x + 1}{(x - 2)^2}$$

MONOTONIA: $f'(x) \geq 0$ ME $\frac{x^2 - 4x + 1}{(x - 2)^2} \geq 0$ $x^2 - 4x + 1 \geq 0$

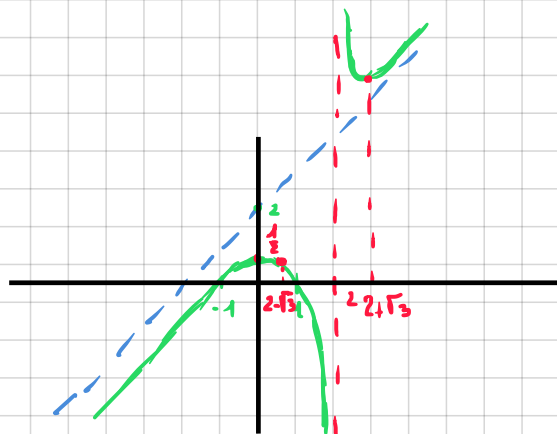
$$\hookrightarrow x_{1,2} = \frac{4 \pm \sqrt{16 - 4}}{2} =$$

$$= \frac{\cancel{2} (2 \pm \sqrt{3})}{\cancel{2}} = 2 \pm \sqrt{3}$$

$$f'(x) > 0 \quad x < 2 - \sqrt{3} \quad \vee \quad x > 2 + \sqrt{3} \quad f'(x) < 0 \quad 2 - \sqrt{3} < x < 2 + \sqrt{3}$$

$$f'(x) = 0 \quad x = 2 \pm \sqrt{3}$$

GRAFICO:



$$f(0) = \frac{0 - 1}{0 - 2} = \frac{1}{2}$$

LIMITI: $\lim_{x \rightarrow 0} \frac{\lim_{x \rightarrow 0} (3x) \arctan(2x)}{x^3 + 2x^2}$

$$\lim_{x \rightarrow 0} (3x) \sim 3x \quad \text{PER } x \rightarrow 0$$

$$\arctan(2x) \sim 2x \quad \text{PER } x \rightarrow 0$$

$$\lim_{x \rightarrow 0} \frac{6x^2}{\cancel{x^2} (x + 2)} = \lim_{x \rightarrow 0} \frac{6}{(x + 2)} = 3$$

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^4 + x}}{x^2 \arctan(x)} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x^4 (1 + \frac{x}{x^4})}}{x^2 \arctan(x)} = \lim_{x \rightarrow +\infty} \frac{\cancel{x^2} \sqrt{1}}{\cancel{x^2} \arctan(x)} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$$

ESERCIZIO 4:

$$a) \int \frac{x+1}{x^3+x^2-2x} dx = \int \frac{x+1}{x(x^2+x-2)} dx = \int \frac{x+1}{x(x+2)(x-1)} dx = (*)$$

$$\frac{x+1}{x(x+2)(x-1)} = \frac{A}{x} + \frac{B}{(x+2)} + \frac{C}{(x-1)} = \frac{A(x^2+x-2) + B(x^2-x) + C(x^2+2x)}{x(x+2)(x-1)} =$$

$$= \frac{x^2(A+B+C) + x(A-B+2C) - 2A}{x(x+2)(x-1)}$$

$$\begin{cases} A+B+C=0 \\ A-B+2C=1 \\ -2A=1 \end{cases} \Rightarrow \begin{cases} A=-\frac{1}{2} \\ -\frac{1}{2}+B+C=0 \\ -\frac{1}{2}-B+2C=1 \end{cases} \Rightarrow \begin{cases} A=-\frac{1}{2} \\ C=\frac{2}{3} \\ B=\frac{1}{2}-\frac{2}{3} = \frac{3-4}{6} = -\frac{1}{6} \end{cases}$$

$$(*) \quad -\frac{1}{2} \int \frac{1}{x} dx - \frac{1}{6} \int \frac{1}{(x+2)} dx + \frac{2}{3} \int \frac{1}{(x-1)} dx = -\frac{1}{2} \log|x| - \frac{1}{6} \log|x+2| + \frac{2}{3} \log|x-1| + C$$

$$b) \int (2x-1) \sin(x) dx = (1-2x) \cos(x) - \int -2 \cos(x) dx =$$

$$= (1-2x) \cos(x) + 2 \int \cos(x) dx =$$

$$= (1-2x) \cos(x) + 2 \sin(x) + C =$$

$$= \cos(x) - 2x \cos(x) + 2 \sin(x) + C$$

$$\begin{array}{ccc} & D & \int \\ & 2 \leftarrow (2x-1) & \\ & \swarrow \text{green} & \downarrow \text{red} \\ \sin(x) & \rightarrow & -\cos(x) \end{array}$$